

PRATHAM SHARMA

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EDUCATION

UNIVERSITY OF FLORIDA (GPA: 3.86/4)

Gainesville, Florida, USA

Master of Science in Computer Science and Engineering

Aug 2023 – Dec 2024

Relevant Coursework: Human-Computer Interaction, Software Engineering, Computer Networks, Advanced Data Structures, Computer Vision, Introduction to Data Science, Machine Learning Engineering, NLP and Data Engineering

TECHNICAL SKILLS

- **Programming Languages:** C++, Java, JavaScript, TypeScript, Python, SQL (PostgreSQL, MySQL), MongoDB (NoSQL), C#
- **Technologies:** React, Redux, Next.js, Node.js, Express.js, Django, Tailwind CSS, HTML, CSS, Socket.io, ASP.NET
- **Cloud and DevOps:** Docker, Kubernetes, Git, JIRA, AWS, Azure
- **Data/ML:** Pandas, NumPy, TensorFlow, PyTorch, Scikit-Learn, Neural Networks (RNN, CNN), LLMs, RabbitMQ

WORK HISTORY

Unified Accessibility - Software Engineer | Remote, United States

Feb 2025 – Present

- **Implemented full-stack platform features** using Next.js, Django, and PostgreSQL with strict adherence to WCAG accessibility standards.
- **Engineered a Django-backed, scalable batch-processing core using Celery and a custom AI-driven analysis engine**, orchestrating multi-stage workflows including element extraction, rule filtering, and accessibility analysis improving throughput and reliability.
- **Automated web element extraction and accessibility audits** using Playwright, capturing annotated DOM screenshots and executing rule-based WCAG checks (e.g., alt-text, contrast validation), reducing manual review time by 50%.

TEA Lab, University of Florida - Research Assistant | Gainesville, FL, United States

Jun 2024 – Dec 2024

- **Flight Simulation Optimization:** Enhanced rocket dynamics accuracy by 10% through advanced simulations in Gazebo using ROS2, ROSFlight, and ROSPlane, streamlining validation processes.
- **Automation & Setup Efficiency:** Automated simulation workflows, reducing setup time by 30% and saving over 50+ man-hours across project phases.
- **Predictive Modeling for Research:** Developed and trained an LSTM model for predictive analysis, increasing simulation efficiency by 25% and improving future model readiness.
- **Data-Driven Insights:** Collected and analyzed simulation outputs to support data augmentation, accelerating experimental iterations and improving research reproducibility.

Global Education Ltd - Full Stack Developer | Maharashtra, INDIA (Hybrid)

Jun 2022 – Aug 2023

- **Proctor Quiz Automation:** Developed a scalable proctor quiz system using React.js, Node.js, Express.js, and MongoDB, automating quiz evaluation and cutting manual grading time by 60%, improving test handling efficiency.
- **Profile & Feature Expansion:** Implemented automated profile creation workflows and enhanced various API/UI features with Redux and Express.js, reducing onboarding effort by 30% and speeding up user setup.
- **Secure Authentication with JWT:** Integrated JWT authentication for APIs to ensure secure and seamless user sessions, safeguarding 5,000+ active users and reducing access-related errors by 40%.
- **API Optimization:** Built and optimized RESTful APIs with Node.js and MongoDB, fixing existing bottlenecks and enhancing API response times by 25%, improving system reliability and frontend integration.
- **CI/CD Deployment Automation:** Set up CI/CD pipelines using GitHub Actions for automated builds, testing, and deployments, cutting release cycle times by 50% and minimizing production bugs.

Georgia Institute of Technology - Research Intern | Atlanta, United States

Aug 2022 – Dec 2022

- **Web Security Analysis:** Conducted comparative audits of landing and login pages across 90+ sites; uncovered key usability-related security gaps and informed improvements in secure UX practices.
- **Data Collection Automation & Time Savings:** Automated data extraction using Python (Beautiful Soup, Regex, NumPy), reducing manual research time by 70% and saving over 120+ analyst hours across the project lifecycle.
- **Security Insights & Risk Reduction:** Analyzed third-party dependencies, SSL certificates, headers, and config patterns, uncovering 35%+ misconfigurations across sites contributing to the design of a risk assessment framework for secure web practices.

PROJECTS

Feedback Nexus - [GitHub Repo](#)

- Built **Feedback Nexus**, a full-stack platform for anonymous feedback using **Next.js, MongoDB, and ShadCN**, increasing collection efficiency by **50%** through real-time updates, secure auth (**NextAuth + JWT**), and input validation with **ZOD**.
- Integrated **AI-powered** features like message assistance, tone customization, and trend analysis, alongside visualizations (**Recharts**) and interactive dashboards for real-time insights.
- Designed a scalable backend with **MongoDB Aggregation Pipeline, caching, and secure APIs**; enhanced performance and UX with **Framer Motion and Aceternity UI** for seamless cross-device responsiveness.

Norman PD Incident Data Pipeline and Visualization - [GitHub Repo](#)

- Constructed a **Python-based** data pipeline that reduced manual processing time by **50%** for Norman, Oklahoma Police Department incident reports, extracting events from PDFs and geocoding locations using the Google Maps API.
- Integrated weather data and geospatial analysis, improving predictive accuracy by **25%** and identifying patterns across locations.
- Automated data enrichment and created **6+** visualizations using **Streamlit, Matplotlib, Seaborn, Plotly, and Scikit-learn**, increasing analysis efficiency by **40%** and aiding public safety resource allocation.